

The Hong Kong Polytechnic University

Department of Electronic and Information Engineering

EIE2111 Assignment 2: Advanced Address Book

(Deadline for Submission: Check the course information)

Introduction

This assignment is an enhanced version of the former assignment in which a software-based address book, which can store personal contact information such as names, telephone numbers and e-mail, addresses of individuals. Information can be retrieved by searching the address book content with reference to the input names, telephone numbers or e-mail addresses of the person of interest. **You must complete Assignment 1 before you start working on Assignment 2.**

The following parts to enhance the former assignment:

1. Replace the fixed size array by a linked list. You are required to change the structure of the class `Record` and the searching method in the main program.
2. Validate all inputs from users. If the input is invalid, you are required to report the corresponding error message and prompt to input again.
3. Provide an option to remove a record.
4. All data can be saved into a random access file (e.g., `data.dat`) and loaded from the file.
5. Modify the program such that it can be executed in graphical mode.

Programming Description

Section A: Class (10 marks)

By using Microsoft Visual Studio, design and implement a simple `Record2` class with the following requirements:

1. Create a new project called `Record2`.
2. Use `Record2` as the class name for the library, and create a header file called `Record2.h` and a class file called `Record2.cpp` file to implement the `Record2` class.
3. For private members of the class, it should include:
 - (a) A text string to store the name of a person.
 - (b) Another text string to store the e-mail address (e.g., `abc@def.com`) of the person.
 - (c) An integer variable to store the 8-digit telephone number of the person.
 - (d) A pointer to store the memory address of the next object of a singly linked list. Note that a doubly linked list is also acceptable.
4. For public members of the class, it should include:
 - (a) A pair of `set` & `get` functions to operate on the text string for names.
 - (b) A pair of `set` & `get` functions to operate on the text string for e-mails.
 - (c) A pair of `set` & `get` functions to operate on the integer variable for 8-digit phone numbers.
 - (d) A pair of `set` & `get` functions to operate on the pointer.

Section B: Main Program (80 marks)

1. The text main menu should be changed to GUI, like Figure 1 (Remark: the second option should be “Create person contact information”). (10 marks)

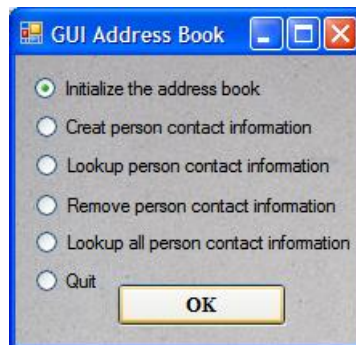


Figure 1 Main menu

2. When the first choice (e.g., initialize the address book) is selected, a message “Initialization is completed.” should be displayed on the screen, like Figure 2. (10 marks)

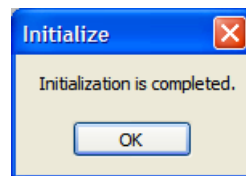


Figure 2

3. When the second choice (i.e., create personal contact information) is selected, the following message should be displayed (see Figure 3) and then ask the end user to input (see Figure 4). Note that, in Figure 4, two buttons “Previous” and “Next” should be provided for the end user to move to the previous and next records respectively. (20 marks)

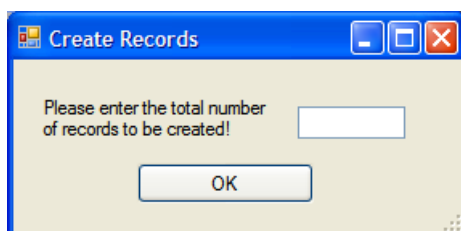


Figure 3



Figure 4

4. When the third choice (i.e., lookup personal contact information) is selected, the following message should be displayed (see Figure 5) and then ask the end user to input. The end user is expected to select a choice and enter the corresponding information. When a record is found, a message “Match Found!” should be displayed on the screen; otherwise, display a message “No Match!”. (20 marks)

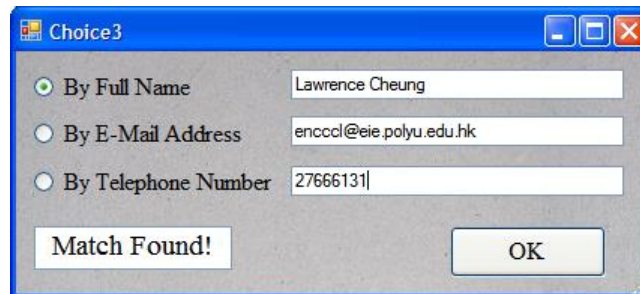


Figure 5

5. When the forth choice (i.e., remove personal contact information) is selected, the following message should be displayed (see Figure 6) and then ask the end user to input. The end user is expected to select a choice and enter the corresponding information. When a record is found, **a message should be displayed to confirm that the record is removed.** (20 marks)

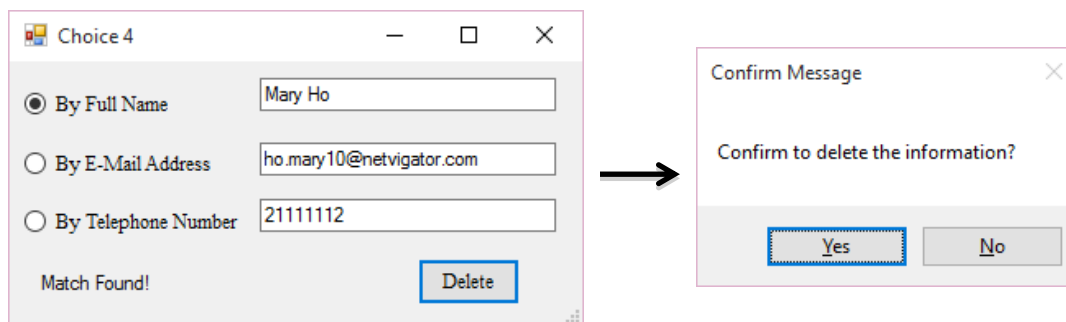


Figure 6

6. When the fifth choice (i.e., lookup all person contact information) is selected, the program should display all person contact information, like Figure 7. (15 marks)



Figure 7

7. If the sixth choice (i.e., Quit) is selected, the program exits. (5 marks)

Section C: The program Flow (10 marks)

Use Microsoft Word or PowerPoint to show the pseudo code or UML activity diagram of the program flow. The design of the program structure is important when assessing your work. It is inappropriate to write all your programs within a single `main()` function.

Instructions

- a. You should zip your programs and the program flow into one file and then submit to Blackboard.
- b. Deadline: **Check the course information.**

Appendix: FAQ

Q: Should I build up a singly or doubly linked list?

A: It is up to you. Both are acceptable.

Q: It is hard to exactly follow the outlook of the demonstration program.

A: The demonstration program is just for your reference. It is not perfect and you can do it better. I post it in this guideline since I want you understand the specification clearly. You are not required to follow the program completely but the specification.

Q: In doing Assignment 2, should I still need to follow all specifications in Assignment 1?

A: Yes, it is because Assignment 2 is the extension of Assignment 1. You should follow all specifications in Assignment 1, except some of them that are asked to be modified in Assignment 2 (e.g., all arrays should be changed to linked lists.). Thus, if your Assignment 1 is not perfect, you should do it again to make it perfect before you do Assignment 2; otherwise, some marks will be deduced in Assignment 2 if it is found that some specifications in Assignment 1 are not exactly followed.

Q: Should I create many functions from DLL to support my C# GUI program?

A: No. It is good enough to create one external function from DLL. In your C# GUI program, you should always call a function (say it `callDLL`) with different input parameters for different functions. Then, in your DLL program, use the input parameters to execute the corresponding function.

Q: It is very inconvenient to modify the program code in my DLL program. What should I do?

A: You should use console program to test all functions before you make them to DLL.

Q: Can I create global variables or a global linked list in my DLL program?

A: No, you cannot. However, you can read the data file, create a linked list to store all data, do whatever you want, then save all data into the file. You do not need to create global variables.